

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1624

Rio Grande City 138 kV
February 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1624. The Interconnection Customer proposes a 122.9 MW battery energy storage facility interconnecting at the Rio Grande City 138 kV substation in the West Virginia area. The request was received on December 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 569 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$786,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: AUBURN RD - PRIDES CORNER 345 KV Line
Contingency: RONCO SUBSTATION - FIRSTENERGY FT MARTIN POWER STA 345 KV line
Loading: 107.1% (Rating: 1,662 MVA)
DFAX: 31.4%
Pre-project loading: 104.8%
- Facility: KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345 KV
Contingency: PRIDES CORNER - MORGANTOWN ENERGY FACILITY 345 KV
Loading: 105.4% (Rating: 1,675 MVA)
DFAX: 5.2%
Pre-project loading: 105.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PRIDES CORNER - AUBURN RD 345 KV	Reconductor line	\$3,430,000	3% (\$104,000)
INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345 KV	Reconductor line	\$27,729,000	2% (\$682,000)
	Total allocated to project		\$786,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.